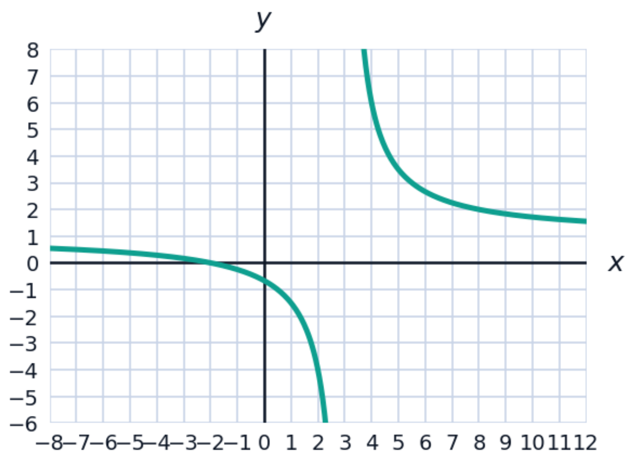
Graphing Rational Functions

Reading graphs

1 The graph of $f(x) = \frac{x+2}{x-3}$ is shown. From the graph and the formula, state the asymptotes and intercepts.



2 The function $g(x) = \frac{1}{x-2} + 3$ is a transformation of $y = \frac{1}{x}$.

a Describe the transformation.

b State the asymptotes of g.

c State the domain and range.

Sign analysis

3 Let $f(x) = \frac{x-1}{(x+2)(x-4)}$. Using a sign chart with critical values $x = -2, 1, 4$, determine the intervals on which $f(x) > 0$.

4 Solve the rational inequality $\frac{x+3}{x-2} \le 0$.

5 Sketch-planning: for $h(x) = \frac{2x}{x^2-1}$, state the asymptotes and the symmetry of the graph.